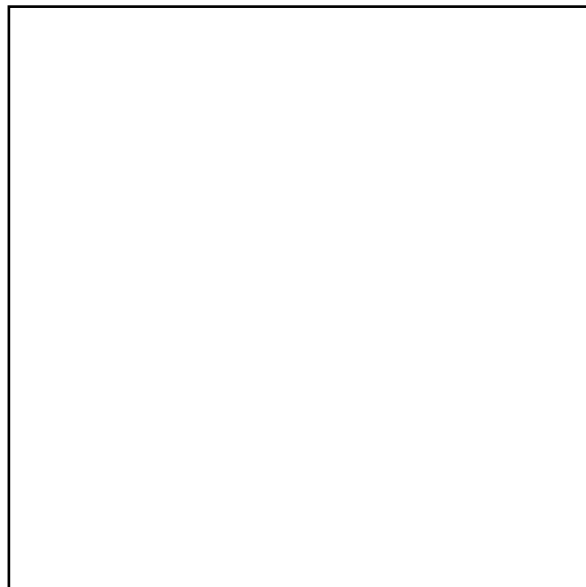


Student number:



EEE3094S 2023: Class Test 1

- The test is open book
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write “see attached pages” by the question) and make sure they are marked with your student number.
- There are four questions totalling 50 marks.
- You have one hour to complete the test.
- The last page is an information sheet.
- The information sheet includes a photo of my cat, Mouse. You are invited to draw a picture of her in the space below if you finish/ give up early and are bored. Doing so will not officially get you extra marks, but it will probably make me and the tutor marking your test happy, which might make a difference in your favour. **Also, the “best” Mouse drawing will gain the artist an unspecified reward.**



Class Test 1

Question 1 [12]

Neuromechanics is an area of biological research that studies how the nervous system controls the musculoskeletal system to bring about movement. A neuromechanics problem that the African Robotics Unit is interested in investigating is the role of the cheetah's tail in balancing its body. Gavin and I recently travelled up to the Ann van Dyk Cheetah Centre to film cheetahs being transported by bakkie. As the car bumps over the rough dirt roads around the sanctuary, the animals often lose their footing and are seen swinging their tails to recover.

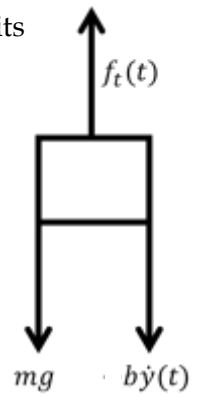
Your task is to draw a block diagram modelling the cheetah in the vehicle. Here is the information your diagram must capture:

- The cheetah can be broken down into the following subsystems: the nervous system $N(s)$, the legs $L(s)$, the tail $T(s)$, the body $B(s)$, the eyes $E(s)$, the vestibular system $V(s)$ and the proprioceptive system $P(s)$.
- The cheetah is trying to control its body angle $\theta(s)$. It uses a combination of the eyes, vestibular system and proprioceptive system to detect this angle. You may consider the sensed angle to be the *average* of these three readings.
- The nervous system considers the difference between the desired angle $r(s)$ and the sensed position, and sends a nervous impulse $i(t)$ to the legs and tail to command them to respond.
- The legs and tail respond to the impulse, producing a combined torque $\tau(s)$ that acts to correct the body angle.
- The motion of the vehicle $c(s)$ also affects the angle of the body.

Question 2 [12]

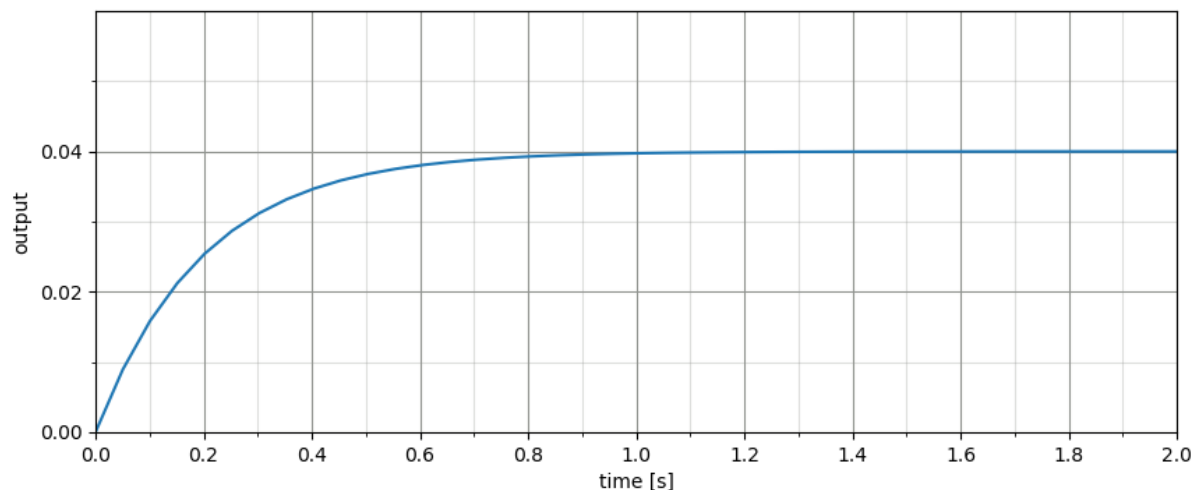
The diagram on the right illustrates the forces acting on a quadcopter. You wish to control its vertical velocity $\dot{y}(t)$.

The rotor thrust $f_t(t)$ is the input to the system. Assuming there is no wind, the aerodynamic drag will be roughly proportional to the velocity of the quadcopter (you can assign it a coefficient b). Gravity acts as a constant external disturbance. The mass of the quadcopter is 1 kg. **You can use $g = 9.8$**



- a) Write a differential equation modelling the quadcopter and hence, show that the relationship between the velocity and rotor thrust can be represented by a first-order transfer function. [2]

- b) You apply a 10 N rotor force to the resting quadcopter, resulting in the change in velocity shown below. Use this to calculate the damping coefficient b and complete the transfer function model of the system. Use *two* features of the given response to confirm your transfer function model of the system. You may annotate the plot to indicate the features of interest. [4]



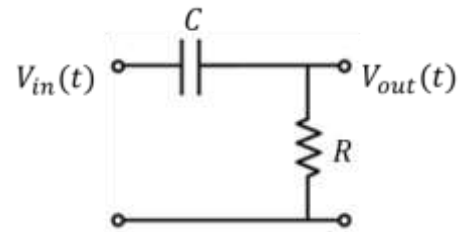
Question 3 [14]

The output voltage of the circuit to the right is related to the input voltage by the differential equation

$$V_{out}(t) + RC \frac{d}{dt} V_{out}(t) = RC \frac{d}{dt} V_{in}(t)$$

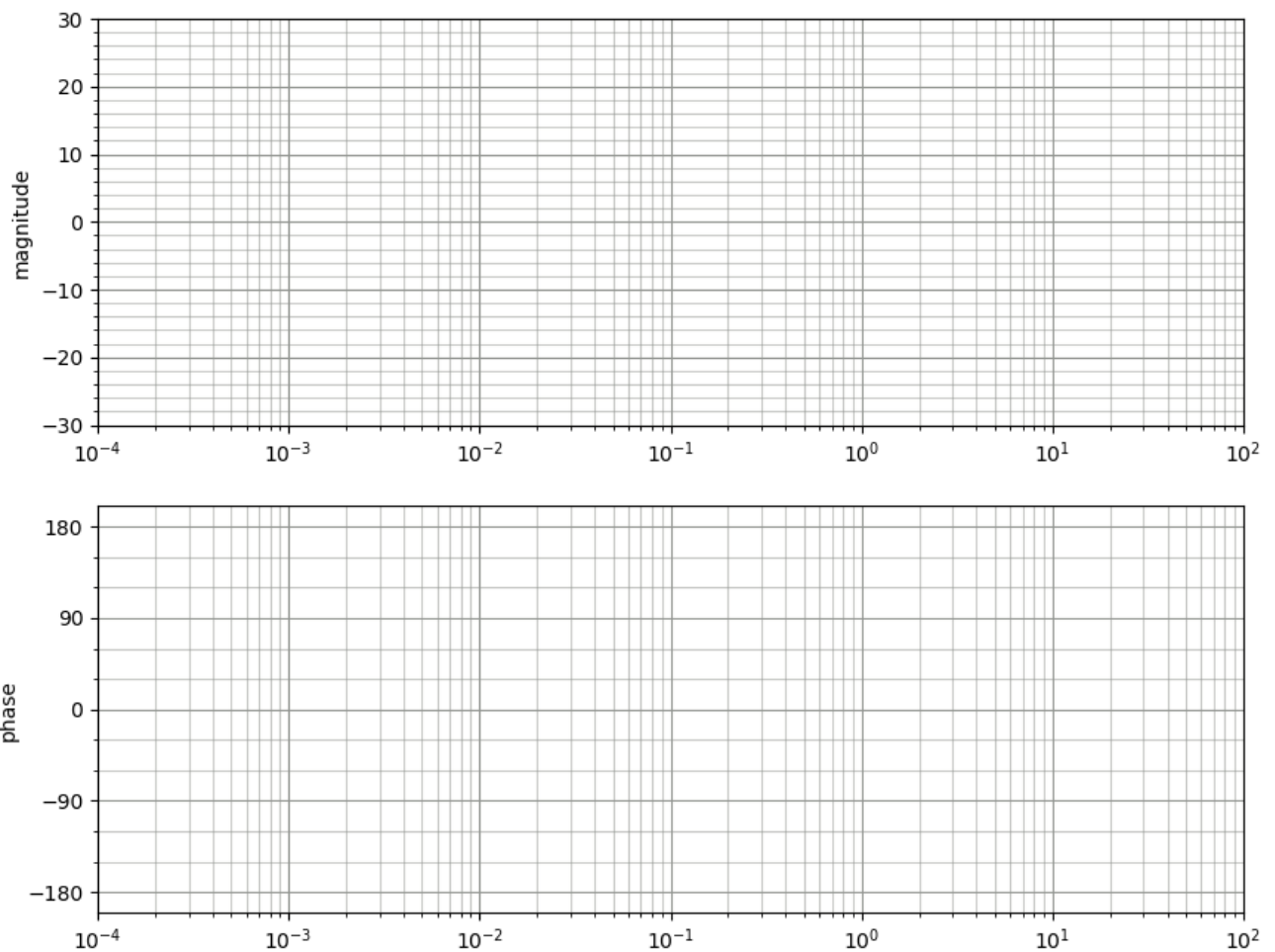
and hence, the transfer function

$$G(s) = \frac{RCs}{1 + RCs}$$



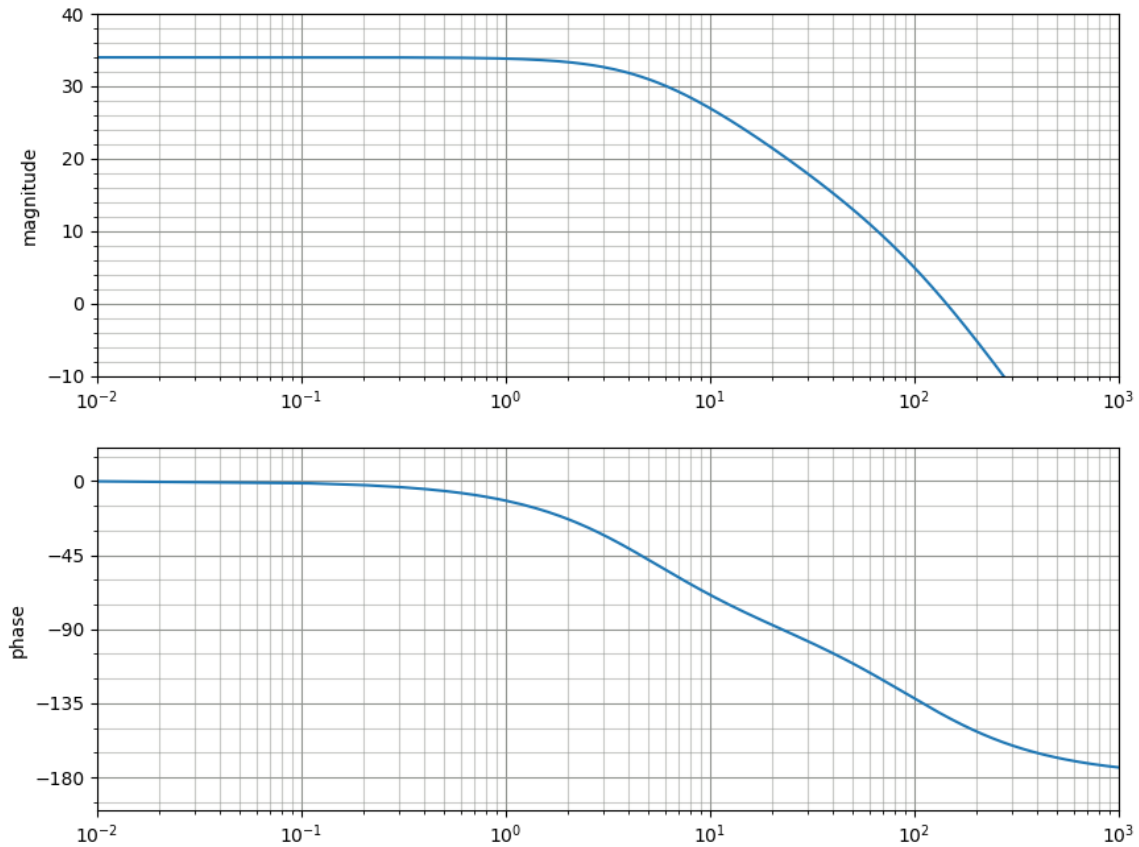
You may use $R = 1 \text{ k}\Omega$, $C = 0.1 \text{ F}$

- a) Roughly sketch the frequency response of this circuit as a Bode plot on the graph below. What type of filter is this? [8]



Question 4 [12]

The Bode plot below gives the frequency response of a second-order system.



- a) The transfer function $G(s) = \frac{50000}{s^2 + 105s + 500}$ is proposed as a model for the system. Confirm this model with reference to the given frequency response. You may indicate relevant features by annotating the plot. [4]

- b) Do you expect that the step response of the system will overshoot? Explain. [2]

Information

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\cos \theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y_p = A \left(1 + e^{-\zeta\pi/\sqrt{1-\zeta^2}} \right)$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln \left(\frac{k}{100} \sqrt{1 - \zeta^2} \right)}{-\zeta\omega_n}$$

Laplace transforms of signals:

$y(t)$	$Y(s)$	Comments
$\sigma(t)$	$\frac{1}{s}$	unit step
$t\sigma(t)$	$\frac{1}{s^2}$	unit ramp
$e^{-at}\sigma(t)$	$\frac{1}{(s+a)}$	decaying exponential if $a > 0$
$t^n e^{-at}\sigma(t)$	$\frac{n!}{(s+a)^{(n+1)}}$	n^{th} order exponential
$\sin(\omega t)\sigma(t)$	$\frac{\omega}{s^2 + \omega^2}$	sinusoid
$\cos(\omega t)\sigma(t)$	$\frac{s}{s^2 + \omega^2}$	<u>cosinusoid</u>
$e^{-at}\sin(\omega t)\sigma(t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$	damped sinusoid
$e^{-at}\cos(\omega t)\sigma(t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$	damped <u>cosinusoid</u>



Fig. 1 Mouse screaming for ice cream